Project Title: RAG-Based Customer Support Assistant

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Course: Internship Project

1. System Overview

This project is a Retrieval-Augmented Generation (RAG)-based Customer Support Assistant.

The system allows users to ask questions related to an e-commerce platform. It retrieves relevant information from a PDF knowledge base and generates accurate answers.

The system also includes a decision-making workflow using LangGraph and supports Human-in-the-Loop (HITL) escalation when confidence is low.

Scope:

- Answer customer queries
- Retrieve information from documents
- Handle low-confidence queries via human escalation

2. Architecture Components

1. User Interface (CLI)
2. Document Loader (PyPDFLoader)
3. Text Splitter (Chunking)
4. Embedding Model (MiniLM)
5. Vector Database (ChromaDB)
6. Retriever
7. LLM (DistilGPT2)
8. Workflow Engine (LangGraph)
9. Routing Logic
10. HITL Module

3. Data Flow

1. PDF is loaded using PyPDFLoader
2. Text is split into chunks
3. Chunks are converted into embeddings
4. Stored in ChromaDB
5. User enters query
6. Retriever fetches relevant chunks
7. LLM generates answer
8. System checks confidence
9. If low → escalate to human
10. Final response returned

4. Technology Choices

ChromaDB:

Used for fast similarity search and efficient storage of embeddings.

LangGraph:

Used to create a graph-based workflow with conditional routing.

Embedding Model:

all-MiniLM-L6-v2 used for lightweight and fast embeddings.

LLM:

DistilGPT2 used for lightweight generation.

Python:

Used for implementation.

5. Scalability Considerations

The system can be scaled by:

- Handling larger PDFs using better chunking strategies
- Using more powerful embedding models
- Deploying vector database on cloud
- Using better LLMs for improved accuracy
- Caching frequent queries to reduce latency